MILESTONE SUBMISSION

Used Cars Price Prediction

Artur Leao de Menezes Lobão

Nowadays there is an extremely high demand for used cars in the Indian market. With the sale of new cars having their speed reduced, not only in the sale but also in the production, we could observe a greater growth in the sale of used and semi-new cars. The company "Cars4U" is entering the market with a new start-up that aims to find footholes in this market.

In 2018-19, while new car sales were recorded at 3.6 million units, around 4 million second-hand cars were bought and sold. There is a slowdown in new car sales and that could mean that the demand is shifting toward the used market. Some car owners replace their old vehicles with used cars instead of buying new automobiles.

Unlike new cars, where price and supply are deterministic and managed by OEMs (Original Equipment Manufacturers / except for dealership level discounts which come into play only in the last stage of the customer journey), the used car market is a quite different beast, with large uncertainties in both pricing and supply. Several factors, including mileage, brand, model, year, etc. can influence the actual worth of a car. From the perspective of a seller, it is not an easy task to set the correct price for a used car. Keeping this in mind, the pricing scheme of these used cars becomes important to grow in the market.

Therefore, using the data tables that were made available to us, I was behind on how to develop the best model with the best variables so that we can explain the best relationship with the value of semi-new or used cars.

Where the key features we find to be able to model the value of cars are:

Name: Name of the car which includes Brand name and Model name

Location: The location in which the car is being sold or is available for purchase (Cities)

Year: Manufacturing year of the car

Kilometers_driven: The total kilometers driven in the car by the previous owner(s) in KM.

Fuel_Type: The type of fuel used by the car (Petrol, Diesel, Electric, CNG, LPG)

Transmission: The type of transmission used by the car (Automatic / Manual)

Owner: Type of ownership

Mileage: The standard mileage offered by the car company in kmpl or km/kg

Engine: The displacement volume of the engine in CC

Power: The maximum power of the engine in bhp

Seats: The number of seats in the car.

From this, if we use trendy belief, the variables that should most affect the value of the car would be:

Model of the car, where usually the most luxurious brands should have a higher value, year of manufacture, where the newer cars should have a higher market value, amount of kilometers driven, where the cars with lower mileage would be the more expensive, transmission, where the automatic cars, are usually more expensive, type of fuel where electric cars would be the most expensive because, in addition to technological woes, they are more modern cars, owners, which represents the number of owners that this car has already owned, where cars with fewer owners, are usually cars that have a lower running and are newer cars and the power of the engines, where usually the most powerful cars, are cars of sportier models and thus more expensive cars.

However, when we perform an analysis of the correlation of these factors with the values of the cars, we can observe that the variables that have this greater correlation are:

Variable	Correlation with the Price variable
Power	0.77
Engine	0.66
Automatic	0.59
Manual	-0.59
Mercedes	0.37

Not Mercedes	-0.37
Diesel	0.32
Not Diesel	-0.32
Year	0.31
Mileage	-0.31
Coimbatore	0.17
Not Coimbatore	-0.17
First Owner	0.09
Not First Owner	-0.09
Seats	0.05
Kilometers driven	-0.01

From this table, we can observe that the variables that have the not variable, have an equal modulo correlation, but with an inverted sign, because they are excluding variables, where if one has a positive correlation, the other necessarily must have the same modulo, but in the different direction. Thus, through the correlation with the price variable, the variables that have the largest module are Power, Engine, Automatic, Manual, Mercedes, Not Mercedes, Diesel, Not Diesel, Year, and Mileage. Here we already see some differences from popular belief and to finish this analysis of variables and create the best model with the most relevant and most significant variables to explain the price variable, we will perform a linear regression between the variables we have and the Price variable, thus arriving at the following variables (where all have a p-value. P>|t|, less than 0.05:

Variable	Coef.	Std.Err	t	P>|t	[0.025	0.975]
Year	0.1055	0.0018	595.717	**0.0000**	0.1020	0.1089
Mileage	-0.0116	0.0016	-71.863	**0.0000**	-0.0148	-0.0084
Engine	0.0002	0.0000	93505	**0.0000**	0.0001	0.0002
Power	0.0056	0.0002	269651	**0.0000**	0.0052	0.0060
Automatic	0.2646	0.0144	183386	**0.0000**	0.2363	0.2929
Coimbatore	0.0893	0.0162	55265	**0.0000**	0.0577	0.1210
First	0.0432	0.0136	31714	**0.0015**	0.0165	0.0699
Diesel	0.2413	0.0127	190125	**0.0000**	0.2164	0.2662
Mercedes	0.1973	0.0237	83406	**0.0000**	0.1510	0.2437

Arriving at the following variables:

Year – the year in which the car was produced, Mileage - The standard mileage offered by the car company in kmpl or km/kg, Engine - The displacement volume of the engine in CC, Power - The maximum power of the engine in bhp, Automatic – Cars with automatic transmission, Coimbatore – cars that are available in the city of Coimbatore, First – Cars that have only one owner, Diesel – Diesel-powered cars and Mercedes brand cars.

With these variables, it was thus possible to use different regression techniques, so that we could arrive at the best pricing model for cars in India.

In my case, I used the following techniques, along with the Gap values between the data that were divided between training and testing, a value that was used as a metric to be able to measure the success of the model, where the lower this GAP, the better the model.

Model Name	Train-Test Gap
Linear Regression	0.0108
Ridge / Lasso Regression	**0.0033**

Decision Trees	0.3497
Decision Trees Hyperparameter	0.0107
Random Forest	0.1957
Random Forest Hyperparameter	0.0224

According to the previous table, we can see that the model that brought us a smaller GAP between training and testing, was the Linear regression model using the Lasso method, with a GAP of 0.0033, thus being considered the best model to describe our problem.

In the future, a way to improve the performance of the models may be, in the Feature Engineering part, to create the variables differently as they were created, in addition to using other parameters in the hyperparameter.